Linear Algebra (MT-1004)

Homework#02

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Question 1

Find the value of t for which the following has a solution

$$\begin{aligned} -w + y - z &= 3 \\ 2w + 2x - y - 7z &= 1 \\ 4w - x - 9y - 5z &= t \\ 3w - x - 8y - 6z &= 1 \end{aligned}$$

Question 2

Using Python's `sympy` library, determine the value of t for which the above system of equations has a solution:

Instructions:

1. Define symbolic variables using `SymPy`.
2. Represent the system of equations symbolically.
3. Use appropriate `SymPy` functions (`linsolve`, `Eq`, etc.) to determine the condition(s) on t that make the system consistent.
4. Print or display the value(s) of t for which the system has at least one solution.